

BRANCHED POLY(ETHYLENEIMINE)-CF₃SO₃Li COMPLEXES

J.-L. PAUL, C. JEGAT and J.-C. LASSÈGUES

Laboratoire de spectroscopie moléculaire et cristalline, Université de Bordeaux I,
351, Cours de la Libération, 33405, Talence Cedex, France

(Received 20 December 1991)

Abstract—Complexes between branched poly(ethylenimine), BPEI, and CF₃SO₃Li have been investigated by *ac* conductivity, calorimetry and *ir* spectroscopy as a function of salt concentration ($4 < N/Li < 25$) and of the temperature (20–100°C). The conductivity presents a maximum of 10^{-6} S cm⁻¹ at 293 K for $N/Li = 20$. Its temperature dependence is of the free-volume type. The N—H groups of this polymer are shown to be very sensitive spectroscopic probes for the *ir* study of the polymer interactions with the cation and the anion. In addition, the evolution of the Li—N stretching vibration in the far-*ir* spectra reveals the occurrence of ion-pairing for $N/Li < 10$.

Key words: lithium conduction, polymer electrolyte, poly(ethylenimine), lithium trifluoromethanesulfonate, infrared spectroscopy.

INTRODUCTION

Poly(ethylenimine) is a widely used polymer because of its interesting properties of complexing metal ions, of acting as a polyelectrolyte *etc.*[1]. We have shown that its protonation by strong acids leads to good proton conducting materials[2, 3]. The high donor number (*dn*) of the amine group (*dn* = 60 for —N— compared to 22 for —O—)[4] has also encouraged several groups to study the alkali ion conduction in poly(ethylenimine)/salt complexes[5–7]. One of the more recent publications in this field emphasizes the interest of using the amorphous branched poly(ethylenimine), BPEI, and indicates a maximum conductivity of 2×10^{-6} S cm⁻¹ at 40°C for the $N/Na = 20$ complex with CF₃SO₃Na[7].

The possible use of such an adhesive polymer as electrolyte in a device working with reversible Li⁺ insertion in electrochromic oxides has lead us to study the complexes of BPEI with a lithium salt.

EXPERIMENTAL

BPEI (Aldrich), mol. wt 5 to 6×10^4 , 50% wt in water, was dehydrated during two days by heating at about 60°C under vacuum. CF₃SO₃Li (Aldrich) was dried under vacuum at 150°C for more than 72 h. Polymer-salt complexes with N/Li from 4/1 to 25/1 were prepared by dissolving stoichiometric quantities of polymer and salt in methanol (Merck) (<0.03% water). Then, the solution was stirred until it was homogeneous. Poly(ethylene glycol) (Aldrich), average mol wt 400, was kept on molecular sieves (<200 ppm water). It was mixed under stirring to the previous alcoholic solution. The percentage $x = EO/(EO + EI)$ of (CH₂CH₂O) repeat units was varied between 0 and 50%. The methanol was removed under vacuum and the complexes were dried at 60°C for 72 h and then kept in a dry box.

Differential Scanning Calorimetry (DSC) measurements were performed on a Mettler TA 3000 system. Glass transition temperatures were measured at a rate of 10 K min⁻¹. The samples were prepared directly from the anhydrous polymer-salt complexes inside the dry box.

For the conductivity measurements, the mixtures were put inside an ion-blocking cell made of stainless steel electrodes previously described[3]. This cell allows the sample to be submitted to a final degassing and drying before enclosing it under vacuum between the two electrodes. The conductivities were measured with a Solartron 1250 frequency response analyser in the range of 1 to 65,000 Hz.

The infrared spectra were recorded with a Nicolet 740 FTIR spectrometer between 4000 and 400 cm⁻¹ and with a Nicolet 20F FTIR spectrometer between 650 and 50 cm⁻¹. The resolution was 4 cm⁻¹ in both cases. For the mid-*ir* spectra, alcoholic solutions were spread on a silicon wafer and evaporated *in situ* whereas for the far-*ir* a thin film was prepared between two poly(ethylene) windows inside the dry-box.

RESULTS AND DISCUSSION

The glass transition temperature T_g of pure BPEI (–47°C) increases by addition of salt as indicated in Table 1, in a very similar way to that reported for the sodium salt[7]. The complexes are amorphous at low salt concentration but the existence of a semi-crystalline phase is indicated, for $N/Li = 4$, by an endotherm peak of melting at 49°C.

Some conductivity data are reported in the $\log \sigma$ vs $1/T$ diagram of Fig. 1. All the samples with $N/Li > 8$ present the non-Arrhenius behaviour already met with the sodium salt[7] and with poly(ether)/salt complexes [4]. This behaviour is characteristic of amorphous phases conductivity and has been interpreted, for instance, in terms of the VTF empirical equation

Table 1. Glass transition temperature T_g and conductivity values of BPEI/CF₃SO₃Li complexes (some conductivity values for a mixture with $x = 50\%$ PEG are given in the last column)

N/Li	BPEI/CF ₃ SO ₃ Li			+ 50% PEG
	$T_g/^\circ\text{C}$	$\sigma^{20^\circ\text{C}}/\text{S cm}^{-1}$	$\sigma^{60^\circ\text{C}}/\text{S cm}^{-1}$	$\sigma^{60^\circ\text{C}}/\text{S cm}^{-1}$
∞	-47	$< 10^{-8}$	—	—
25	-36	$7.4 \cdot 10^{-7}$	$1.4 \cdot 10^{-5}$	—
20	-31	10^{-6}	$3.3 \cdot 10^{-5}$	$6.4 \cdot 10^{-5}$
16	-27	$4.5 \cdot 10^{-7}$	$3.4 \cdot 10^{-5}$	—
8	-5	—	$8.5 \cdot 10^{-7}$	$9.4 \cdot 10^{-5}$
4	—	—	—	$1.4 \cdot 10^{-5}$

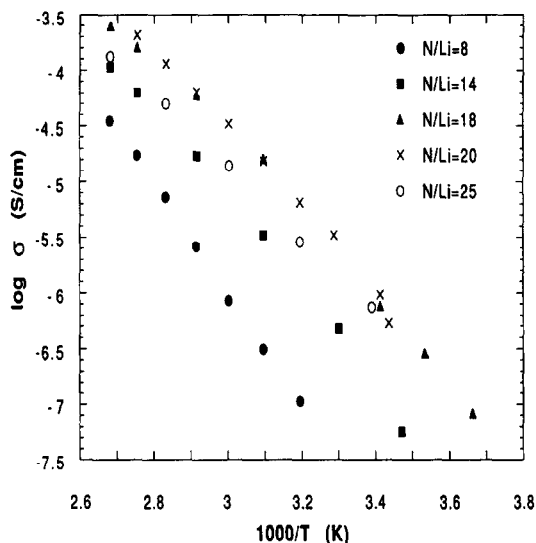


Fig. 1. Arrhenius plot of the conductivity of BPEI-CF₃SO₃Li complexes.

$\sigma = AT^{-1/2} \exp[-E_a/k_B(T - T_0)]$, where A is proportional to the charge carrier concentration, E_a is the activation energy and T_0 the ideal glass transition temperature. These parameters were fitted to give the values reported in Table 2.

Thus, the conductivity increases when the salt concentration and the T_g values decrease. A maximum is obtained for a low concentration of salt near $N/Li = 20$. Addition of PEG shifts this maximum to much higher salt concentrations and improves the conductivity (Table 1) as expected from the double action of PEG in plasticizing the polymer and solvating the cations.

Infrared spectroscopy has been used to characterize the polymer-salt interactions. Only some specific spectral regions are commented here to illustrate the method.

Table 2. VTF parameters for various BPEI/CF₃SO₃Li compositions (the ratio T_g/T_0 has been found to be very close to 1.3 in all cases)

N/Li	$A/\text{S} \cdot \text{cm}^{-1} \cdot \text{K}^{1/2}$	E/eV	T_0/K
8	71	0.166	206.3
14	26	0.149	189.9
16	19	0.132	189.4
18	14	0.130	183.2
25	13	0.141	182.2

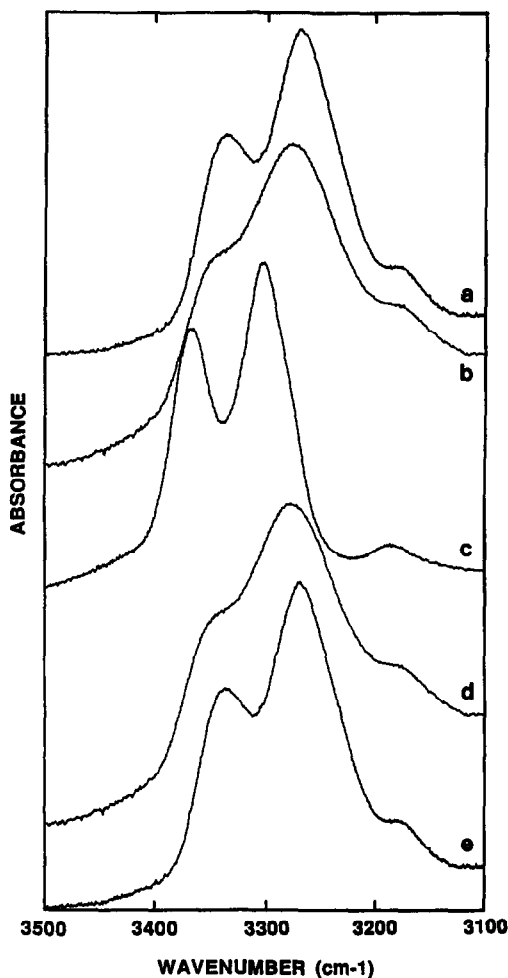


Fig. 2. *ir* spectra of the stretching vibrations of the BPEI amine groups: for pure BPEI (a), BPEI-CF₃SO₃Li complexes with $N/Li = 20/1$ (b), $4/1$ (c), and BPEI/SCNLi complexes with $N/Li = 20/1$ (d), $4/1$ (e).

The stretching vibrations of the amine groups are represented in Fig. 2 and the ν_{as} and ν_s wavenumbers of the $-\text{NH}_2$ groups are reported in Fig. 3. The $\text{N}-\text{H} \dots \text{N}$ hydrogen bonding interactions in pure BPEI are evidenced by a decrease of about 60 cm^{-1} of both ν_{as} and ν_s from the values of, for example, gaseous ethylenediamine[8]. When CF₃SO₃Li is dissolved in BPEI, the ν_{as} and ν_s vibrations occur at slightly higher wavenumbers than in the pure polymer as a result of the progressive substitution of the $\text{N}-\text{H} \dots \text{N}$ hydrogen bonds by $\text{Li}^+ \dots \text{N} \dots \text{O}_3\text{S}-\text{CF}_3$ interactions. It is well known that both cation-nitrogen coordination and $\text{N}-\text{H} \dots \text{anion}$ hydrogen bond contribute to a weakening of the $\text{N}-\text{H}$ bond and therefore to a lowering of its gas phase stretching frequency[9]. As indicated in Fig. 3, the global effect is of about the same order of magnitude as that of the $\text{N}-\text{H} \dots \text{N}$ interactions in pure BPEI. However, secondary effects can also be evidenced: an increase of the salt concentration produces shifts to higher wavenumbers of up to 10 cm^{-1} (Fig. 2). Then, for a given salt concentration, substitution of Li^+ by Na^+ increases ν_{as} by 15 cm^{-1} and ν_s by 3 cm^{-1} (Fig. 3) in agreement

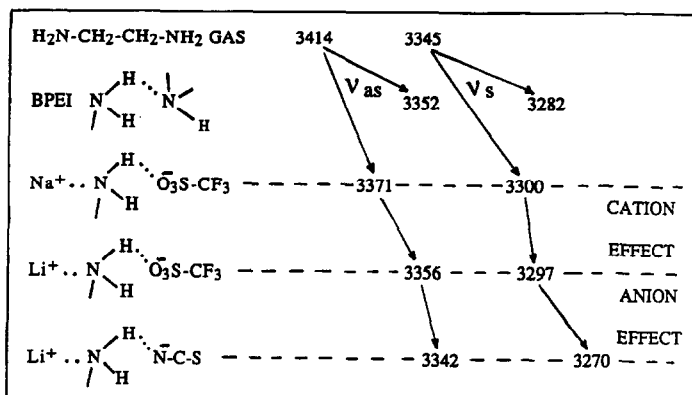


Fig. 3. Schematic representation of the ν_{as} and ν_s positions (cm^{-1}) from the gas phase reference value of ethylenediamine[8] to pure BPEI and to its complexes with sodium[7] or lithium salts involving N/Li and N/Na = 12/1.

with a Na-N interaction weaker than the Li-N one. On the other hand, substitution of CF₃SO₃⁻ by SCN⁻ decreases ν_{as} by 14 cm^{-1} and ν_s by 27 cm^{-1} (Fig. 3) because the hydrogen bond interaction is stronger with SCN⁻.

The major conclusion from the above observations is that a polyamine is much more favourable than for example a polyether for study of the polymer-salt complexation by *ir* spectroscopy simply because the N-H oscillators are very sensitive probes for the cation-polymer and anion-polymer interactions. A more fundamental consequence of these latter interactions might be a smaller anionic transport number in polyamines than in polyethers.

Of course, other internal modes of the polymer exhibit intensity and frequency changes upon complexation. They can be used to complement the previous analysis and in particular to follow conformational changes. However, we have represented in Fig. 4 another spectral region of interest for any polymer-salt complex: the observed broad absorption is due to the lithium stretching vibration in the field of the solvating ligands. It is centred at about 480 cm^{-1} in

diluted BPEI/CF₃SO₃Li mixtures and it is shifted to 180 cm^{-1} when Li is replaced by Na. For N/Li < 10, this broad absorption becomes relatively more intense, is displaced to lower wavenumbers and becomes asymmetrical. This can be interpreted in terms of increasing ion-pairing, the extreme limit being given by the spectrum of the pure salt (Fig. 4). This evolution is accompanied by a marked broadening of several internal modes of the anionic part.

CONCLUSION

The conductivity and calorimetric data we have obtained for the BPEI-CF₃SO₃Li complexes are very similar to those previously published by Harris *et al.* [7] with the sodium salt. In particular, the best conductivity is obtained for low salt contents (N/Li = 20) and is only 10⁻⁶ S cm^{-1} at 293 K. However, the adhesive and amorphous character of BPEI and the possibility of increasing the conductivity with a plasticizer, provide hope for possible applications of the BPEI/CF₃SO₃Li system as a Li⁺ conducting electrolyte.

REFERENCES

1. *Polymeric Amines and Ammonium Salts* (Edited by E. J. Goethals). Pergamon Press, Oxford (1979).
2. M. F. Daniel, B. Desbat, F. Cruege, O. Trinquet and J. C. Lassègues, *Solid St. Ionics*, **28-30**, 637 (1988).
3. J. C. Lassègues, D. Schoolmann and O. Trinquet, Symposium A2-Solid State Ionics of the *Int. Conf. Adv. Mat. (ICAM 1991)*, Strasbourg, France, 27-31 May 1991 (Edited by M. Balkanski), pp. 443-448. Elsevier, Amsterdam (1991).
4. M. B. Armand, *Solid St. Ionics* **9,10**, 745 (1983).
5. T. Takahashi, G. T. Davis, C. K. Chiang and C. A. Harding, *Solid St. Ionics* **18,19**, 321 (1986).
6. C. S. Harris, D. F. Shriver and M. A. Ratner, *Macromol.* **19**, 987 (1986).
7. C. S. Harris, M. A. Ratner and D. F. Shriver, *Macromol.* **20**, 1178 (1987).
8. A. Sabatini and S. Califano, *Spectrochim. Acta* **16**, 677 (1960).
9. K. Nakamoto, *Infrared and Raman Spectra of Inorganic and Coordination Compounds*. 4th Edn, p. 191. John Wiley, New York (1986).

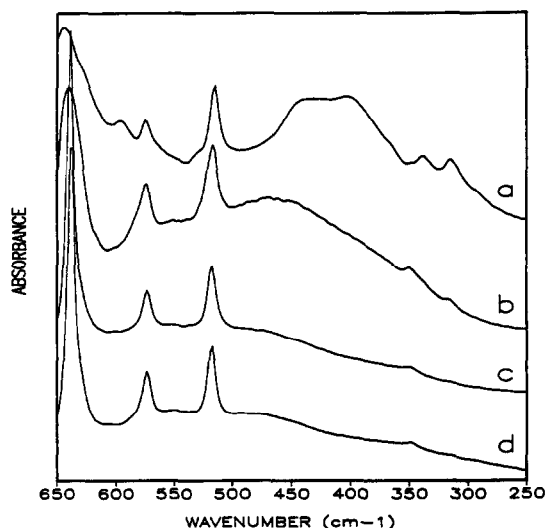


Fig. 4. Far-*ir* spectra of CF₃SO₃Li(a) and of BPEI-CF₃SO₃Li complexes with N/Li = 4/1(b), 16/1(c), 20/1(d).